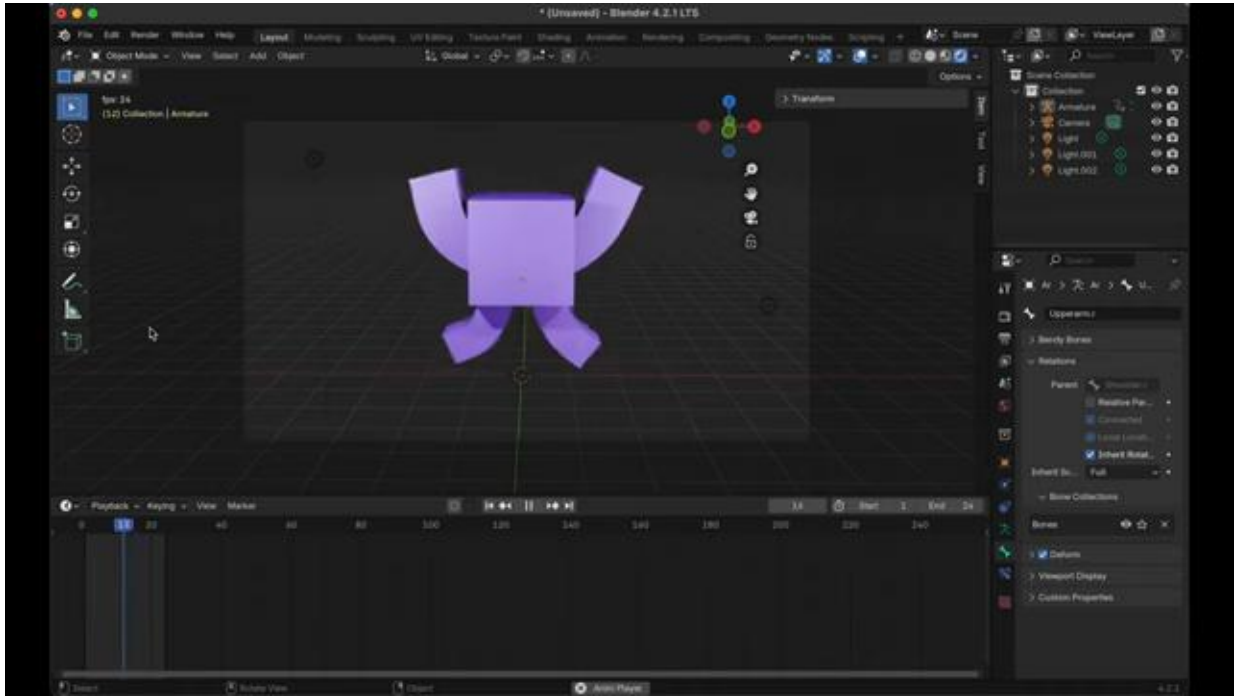
3D Creative Club Study Guide

Armature (Bones)

Goal: What You Will Build

Object Mode



Steps:

- Build a simple box man from cubes
- Create an armature (bone skeleton) inside it
- Bind the mesh to the bones so they control the body
- Test the rig in Pose Mode
- (Optional) Texture paint, then animate and share on Teams

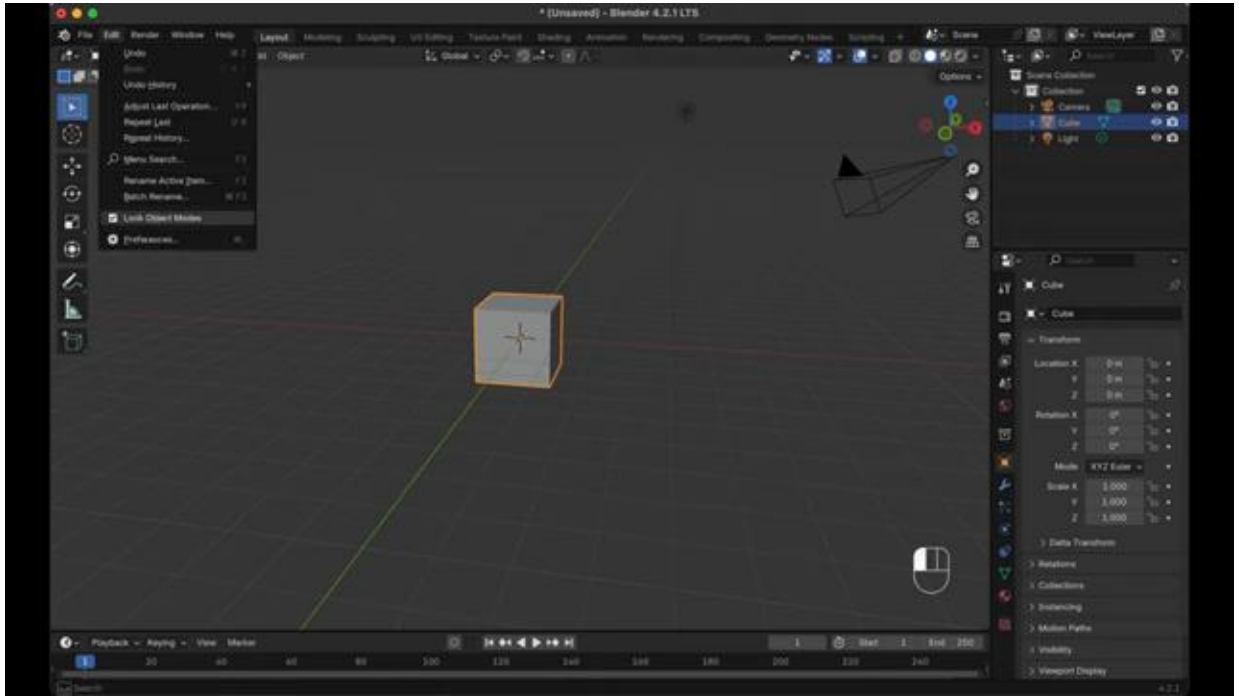
Shortcuts:

Shift + A: Add · Tab: Mode toggle · Ctrl + P: Parent / bind

Tip: Last week you animated a pre-rigged model. This week you BUILD the rig yourself.

Step 1: Model the Box Man

Edit Mode



Steps:

- Start with one Cube (or press X to delete the default cube, then Shift + A → Cube)
- Add more cubes for head, torso, arms, and legs
- Scale (S) and move (G) each part into a simple human shape
- Use G/S + X/Y/Z to align along one axis

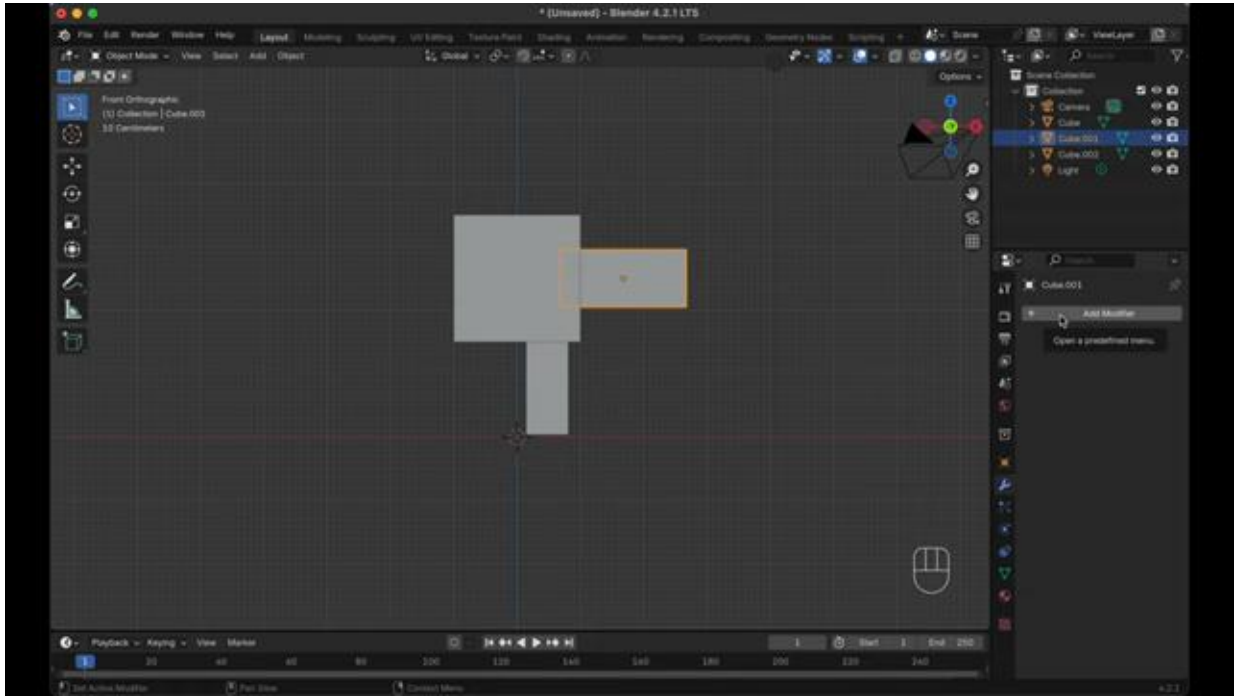
Shortcuts:

Shift + A: Add · G/S: Move / Scale · G/S + X/Y/Z: One axis

Tip: Keep it blocky — a box man does not need fingers or facial features.

Step 2: Join & Merge Parts

Object Mode



Steps:

- In Object Mode, select all body parts
- Press Ctrl + J to join them into one mesh
- Switch to Edit Mode (Tab)
- Select all vertices and press M → Merge By Distance to remove overlapping vertices

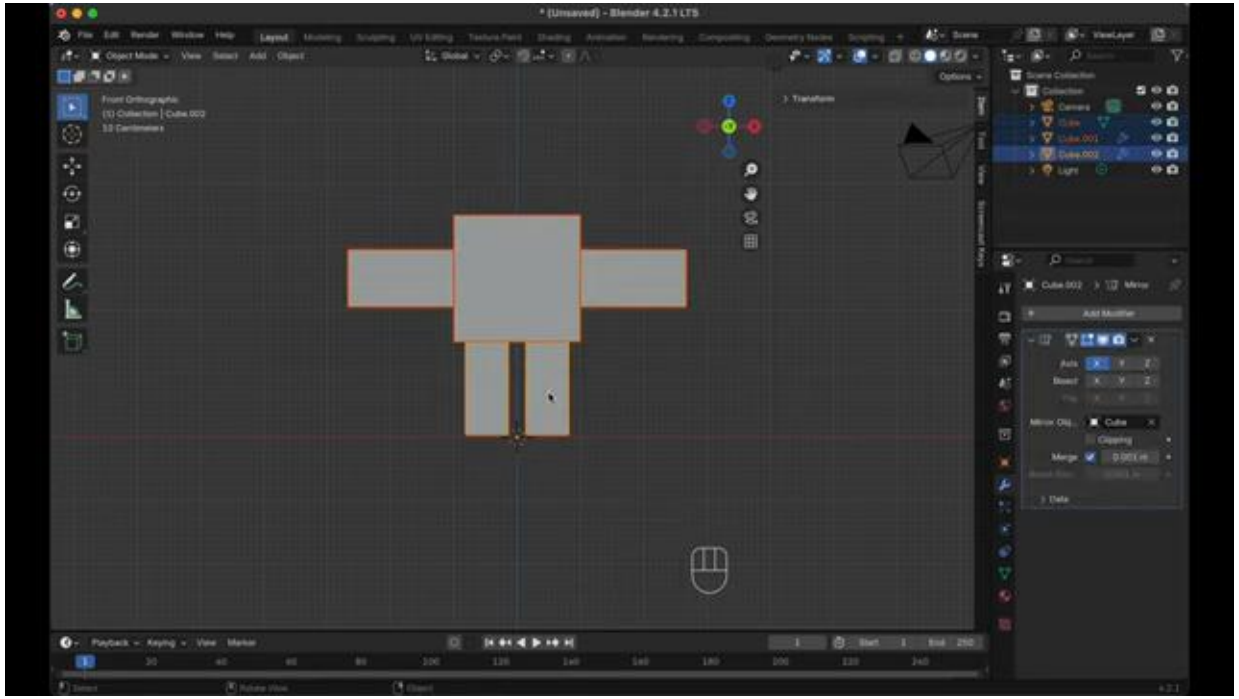
Shortcuts:

Ctrl + J: Join meshes · A: Select all · M: Merge

Tip: Overlapping vertices can break rigging, so this cleanup step is important.

Step 3: Add Joint Loop Cuts

Edit Mode



Steps:

- Stay in Edit Mode on the joined mesh
- Press Ctrl + R to add loop cuts at shoulders, elbows, hips, and knees
- Left-click to confirm, then move the cut to the joint line
- Add more cuts if an area needs to bend smoothly

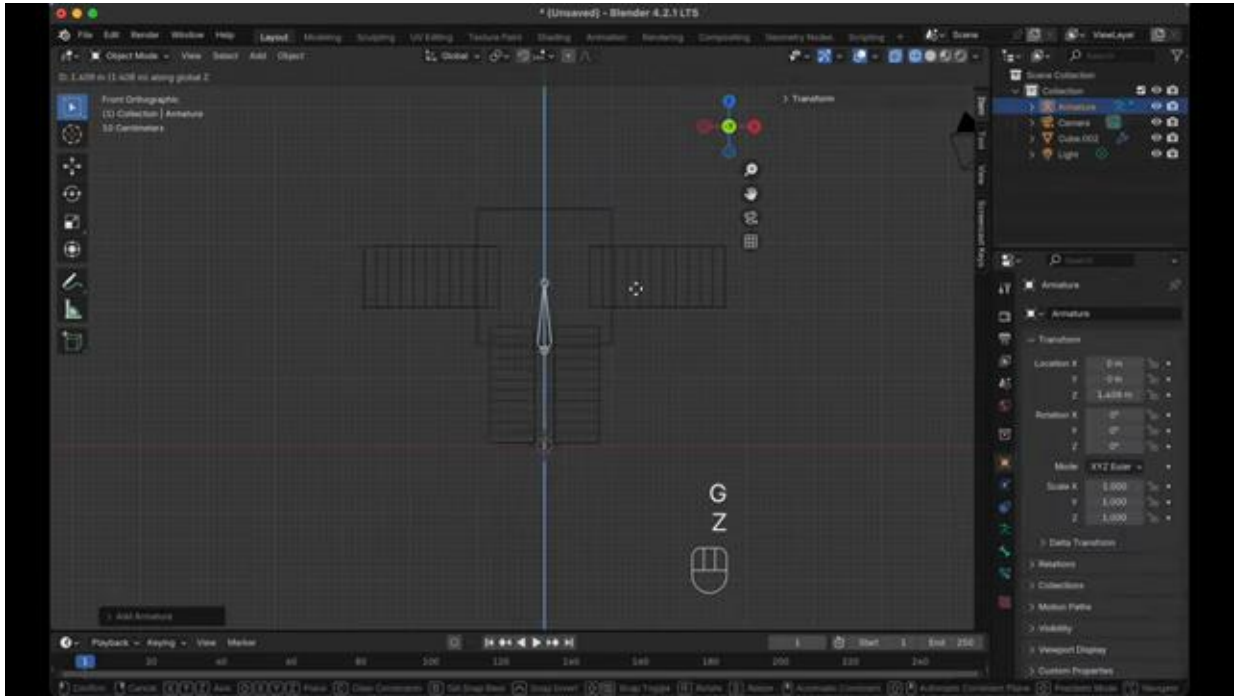
Shortcuts:

Ctrl + R: Loop cut · Wheel: More cuts · Left click: Confirm

Tip: Loop cuts give the mesh extra geometry so it can bend at joints.

Step 4: Create the Armature

Armature Edit Mode



Steps:

- Switch to Object Mode, then Shift + A → Armature → Single Bone
- Press Tab to enter Armature Edit Mode
- Select the tip of the bone and press E to extrude new bones
- Build a spine chain, then extrude head, arms, and legs

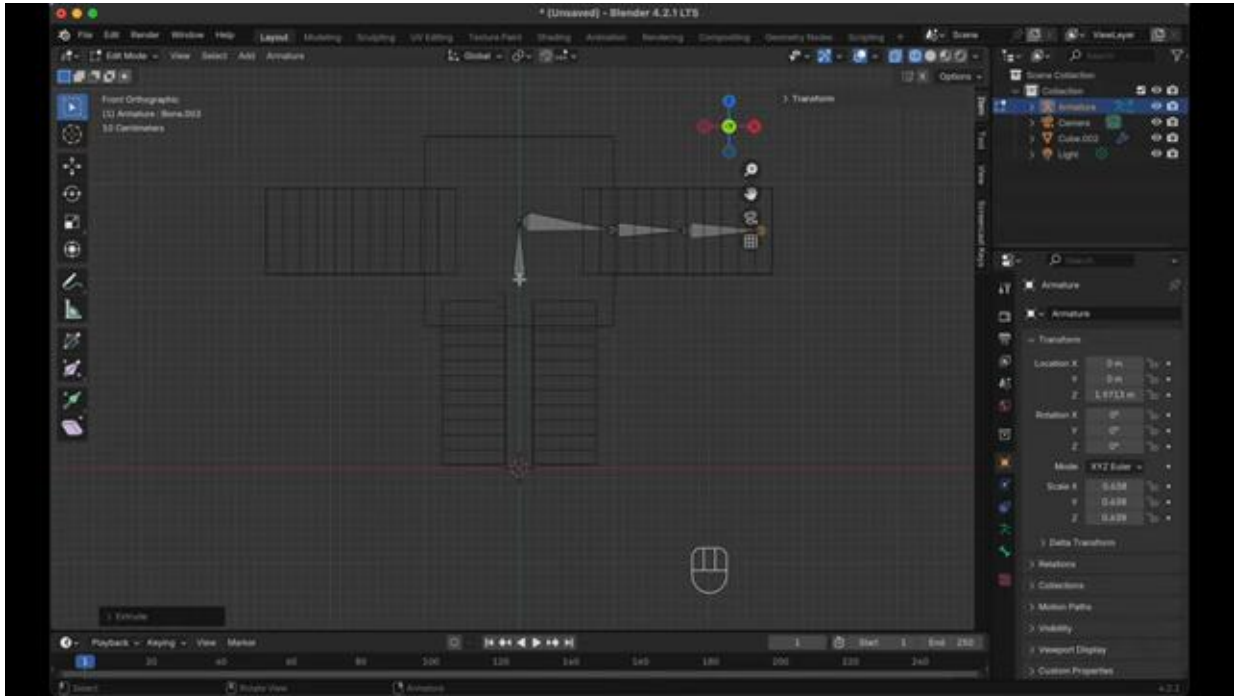
Shortcuts:

Shift + A: Add armature · Tab: Edit Mode · E: Extrude bone

Tip: Make sure the bones are placed inside the mesh, not floating outside.

Step 5: Parent Bones & Build Hierarchy

Armature Edit Mode



Steps:

- Select a child bone, then Shift-select its parent bone
- Press Ctrl + P → Connected (or Keep Offset)
- Build this chain: spine → head, spine → arms, hips → legs
- Check the Outliner to see the bone hierarchy

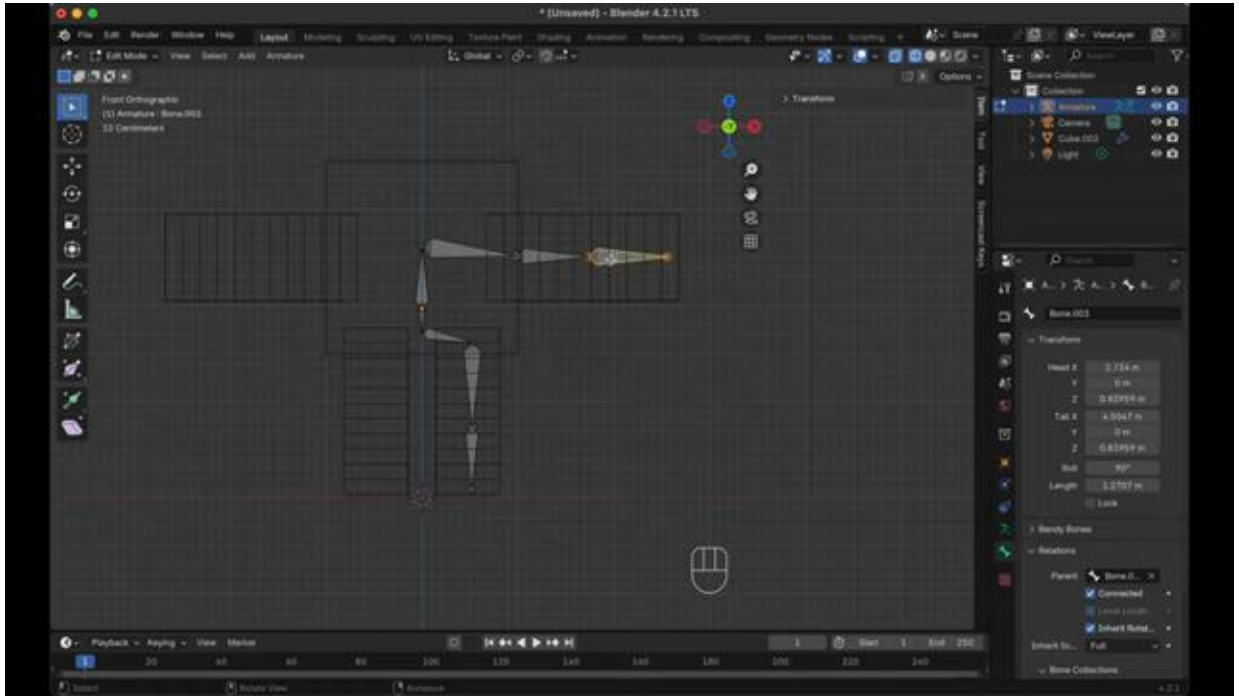
Shortcuts:

Ctrl + P: Parent bones · Shift + click: Multi-select

Tip: Hierarchy matters: moving the spine should move the whole upper body.

Step 6: Name & Symmetrize Bones

Armature Edit Mode



Steps:

- Name left-side bones with .L suffix, e.g., arm.L, leg.L
- In Edit Mode, right-click the left bone
- Choose Armature → Symmetrize
- Blender automatically creates the matching .R bones on the other side

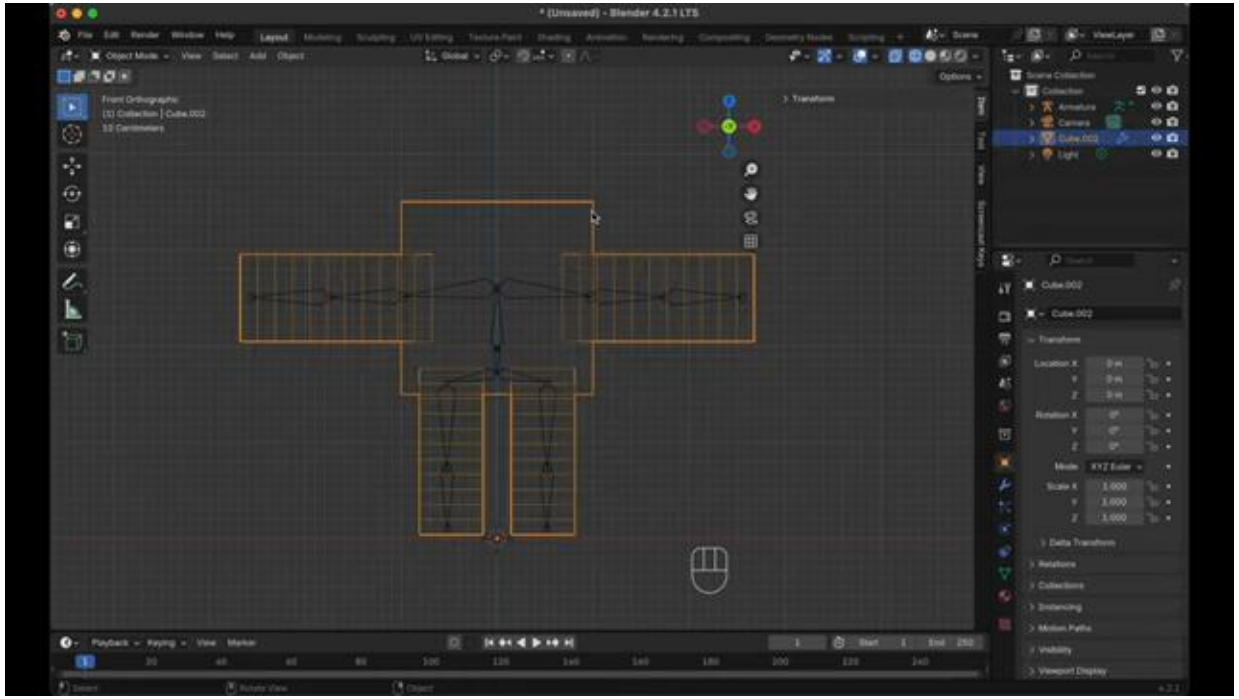
Shortcuts:

F2: Rename · Right click → Symmetrize

Tip: The .L / .R suffix is required for Symmetrize to work correctly.

Step 7: Bind Mesh to Armature

Object Mode



Steps:

- Switch to Object Mode
- Select the mesh FIRST, then Shift-select the Armature
- Press Ctrl + P → Armature Deform → With Automatic Weights
- The mesh is now controlled by the bones

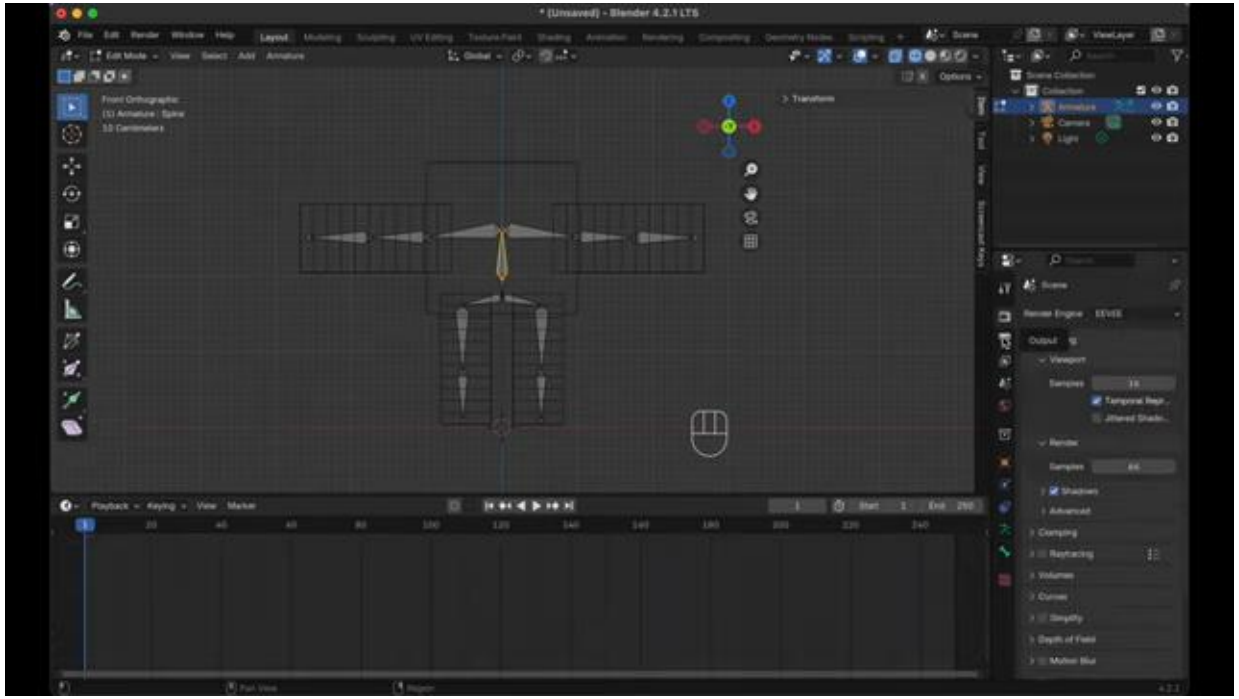
Shortcuts:

Ctrl + P → Armature Deform → With Automatic Weights

Tip: Selection order matters: mesh first, armature second.

Step 8: Test Your Rig in Pose Mode

Pose Mode



Steps:

- Select the Armature, press Ctrl + Tab to enter Pose Mode
- Click a bone and press R to rotate it
- The mesh should move with the bone
- If a part bends wrong, the weights may need fixing (Weight Paint mode)

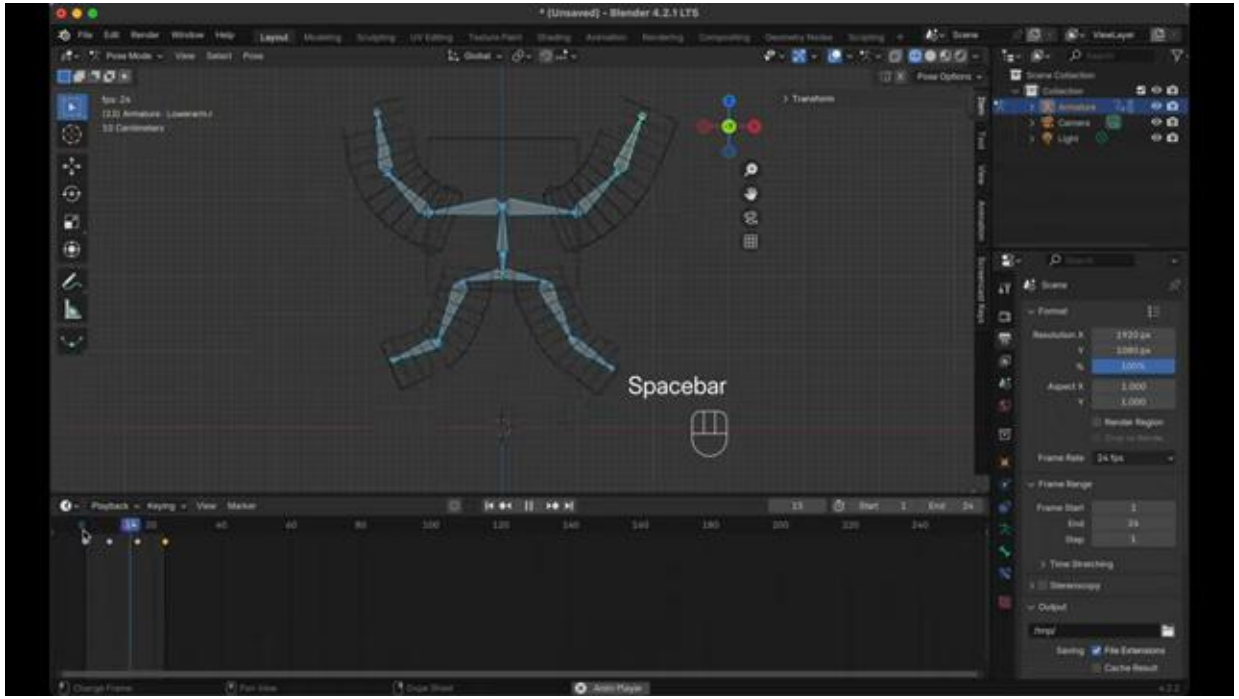
Shortcuts:

Ctrl + Tab: Pose Mode · R: Rotate bone · Tab: Exit Pose Mode

Tip: If the rig works, you are ready to animate using last week's keyframe skills.

Optional: Texture Paint

Optional

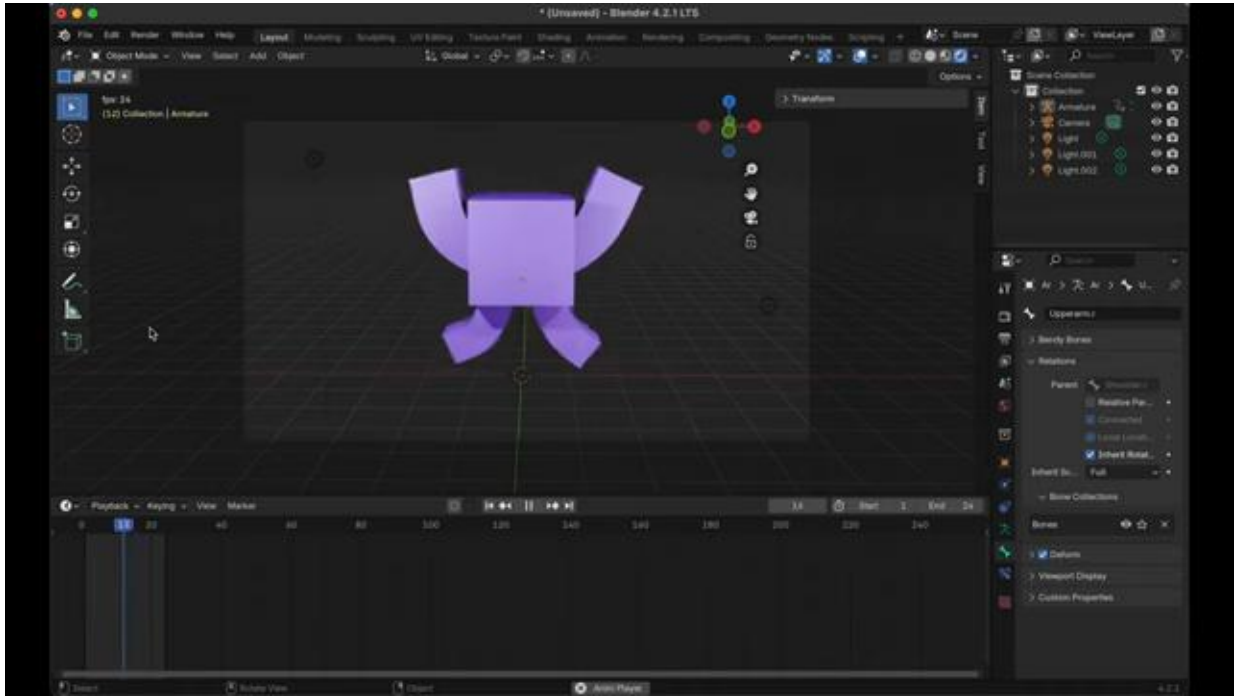


What to do:

- Finished early? Add color to your box man.
- Switch to the Texture Paint workspace.
- Or use Material Properties → Base Color to set flat colors.
- Paint the body, shirt, pants, and face directly on the model.

Tip: This is not required today, but it makes your final render look much better.

Animation Challenge & Peer Review



What to do:

- Use last week's keyframe skills: Pose Mode → I to insert keyframes.
- Create a short animation with your own rigged box man.
- Upload your video to Teams for peer review.
- Like your classmates' work — most likes wins a 3D-printed prize next class!

Tip: Save your rendered video to a USB or cloud drive — shared computers may delete local files.

Shortcut Cheat Sheet

Tab	Toggle Edit / Object Mode	**Ctrl + Tab**	Toggle Pose Mode	**Shift + A**	Add object / bone
Shift + D	Duplicate	**G / S / R**	Move / Scale / Rotate	**G/S/R + X/Y/Z**	Constrain to axis
Ctrl + J	Join meshes	**Ctrl + R**	Loop cut	**E**	Extrude face / bone
Ctrl + P	Set parent / bind	**M**	Merge vertices	**F3**	Search menu